

CONFIGURABLE-BANDWIDTH TIME-FREQUENCY MODELING FOR EFFICIENT FULL-BAND SPEECH ENHANCEMENT ACROSS SAMPLING RATES

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ABSTRACT

Speech enhancement systems are often developed for a fixed sampling rate, while time-frequency models become more expensive as the number of frequency bins increases. We propose *TF-Refiner*, a sampling-frequency-independent model that decouples the deep analysis bandwidth from the full-band input and output. A deep encoder processes the band below a configurable cutoff, while a shallow decoder combines the encoded features with input-dependent high-band queries and predicts local complex filters applied to the original noisy STFT. A single parameter set trained at 16 and 48 kHz is evaluated at various sampling rates. On Voice-Bank+DEMAND, the *universal* model outperforms the rate-specific counterparts in PESQ, STOI, and log-spectral distance across the evaluated rates, including rates unseen in training. Random-cutoff training enables inference-time selection of cost-quality operating points without retraining or changing the output bandwidth. These results support configurable analysis bandwidth as a practical design choice for multi-rate full-band enhancement.

Index Terms— speech enhancement, full-band audio, sampling-frequency independence, configurable computation

1. INTRODUCTION

Speech enhancement (SE) is deployed at more than one sampling rate: telephony and on-device pipelines operate at 16 kHz, whereas conferencing and media applications increasingly require full-band 48-kHz audio [1], whose band above 8 kHz carries cues of naturalness and intelligibility [2, 3, 4]. In practice, a separate model is typically trained for each rate: lightweight streaming models [5, 6, 7] for 16 kHz and full-band systems [8, 9, 10, 11, 12] for 48 kHz. Sampling-frequency-independent (SFI) processing removes this duplication in principle by tying the network to physical frequency rather than to a fixed bin index [13, 14, 15], and time-frequency (TF) dual-path models [16, 17, 18], which treat the frequency axis as a sequence, support this representation naturally. However, existing SFI enhancement systems have relatively high computational demands [15, 19, 20], and the cost of a TF dual-path model grows almost proportionally with the number of bins, so that the same

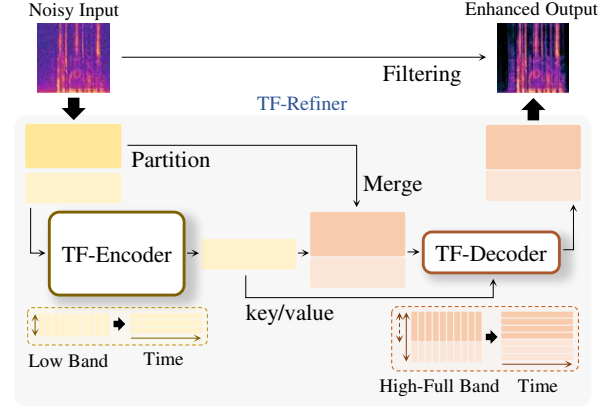


Fig. 1. Overall flow of TF-Refiner: deep analysis of the band below f_c , shallow full-band decoding with cross-attention to the encoder output, and filtering of the original noisy STFT.

architecture becomes substantially more expensive at 48 than at 16 kHz [21].

Practical full-band systems avoid this growth with hand-designed frequency structures, such as perceptual-band envelopes [8, 9], equivalent-rectangular-bandwidth (ERB) gains refined by complex filtering only in the low band [11], or fixed sub-band networks [12, 22]. In these designs, however, the allocation of analysis effort across frequency is fixed at design time and tied to one sampling rate.

These observations motivate a *configurable analysis bandwidth*: a single SFI model whose deep analysis is confined to a band defined in physical frequency, while its full-band input and output follow the sampling rate. To this end, we propose *TF-Refiner*, an asymmetric encoder-decoder for full-band SE, as shown in Figure 1. It builds on the query-based asymmetric structure of TF-Restormer [21], which analyzes only the observed band and synthesizes the unobserved band from learnable extension queries, and moves the band partition from the edge of the observed spectrum to a configurable cutoff f_c inside it. A deep TF-Encoder analyzes the band below f_c , whereas the complementary observed band enters a shallow TF-Decoder as input-dependent queries that attend to the encoder output [23, 24], so that the encoder cost is set by f_c rather than by the sampling rate. Rather than mapping the spectrum directly, the decoder predicts local complex filters applied to the original noisy STFT [25, 26], preserving the observed resolution. Unlike the frequency projection of TF-Restormer, which fixes a maximum bin count, no parameter of TF-Refiner depends on the number of bins, enabling one parameter set to serve any sampling rate.

Experiments show that a single *universal* model trained at 16

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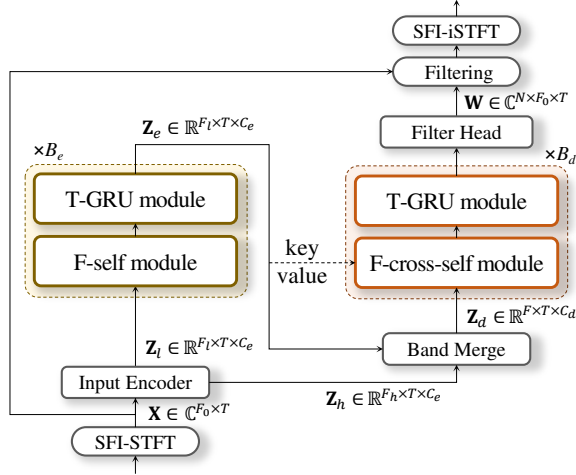


Fig. 2. Architecture of TF-Refiner. The F_l positions below f_c form $\mathbf{Z}_l$ for the deep TF-Encoder and the remaining F_h positions form $\mathbf{Z}_h$. Band Merge concatenates $\mathbf{Z}_h$ with the encoder output $\mathbf{Z}_e$ for the shallow TF-Decoder, which also attends to $\mathbf{Z}_e$ as key/value.

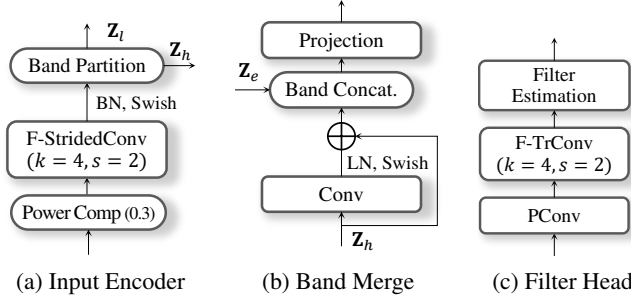


Fig. 3. Components outside the TF modules.

and 48 kHz matches or exceeds rate-specific training in perceptual quality and generalizes to unseen rates, while random-cutoff training lets f_c set the cost-quality trade-off at inference without retraining.

2. TF-REFINER

As shown in Figure 2, TF-Refiner realizes the configurable bandwidth through an SFI representation that keeps f_c at the same physical frequency for every rate, a deep TF-Encoder for the analysis band only, a shallow TF-Decoder that refines the complementary observed band through cross-attention, and a Filter Head that predicts local complex filters for the original noisy STFT. F_l is fixed by f_c , so for a given cutoff the sampling rate enters only through $F_h = F - F_l$.

2.1. SFI representation and band partition

The SFI-STFT [13, 21] keeps a 40-ms square-root Hann window and a 20-ms hop at every sampling rate f_s , so the bin spacing is 25 Hz and the one-sided spectrum $\mathbf{X}$ has $F_0 = 0.02f_s + 1$ bins. The Input Encoder (Figure 3(a)) compresses the magnitude as $\mathbf{X}_c = |\mathbf{X}|^c \cdot e^{j\angle\mathbf{X}}$, where $c = 0.3$, and embeds the spectrum with a frequency stride of 2, giving $\mathbf{Z}$ with $F = (F_0 + 1)/2$. Band Partition splits the grid at the position of f_c into $\mathbf{Z}_l$ and $\mathbf{Z}_h$ by $F_l = 0.02f_c + 1$. We draw f_c at random during training, so that one parameter set serves any cutoff at inference (Section 4.2).

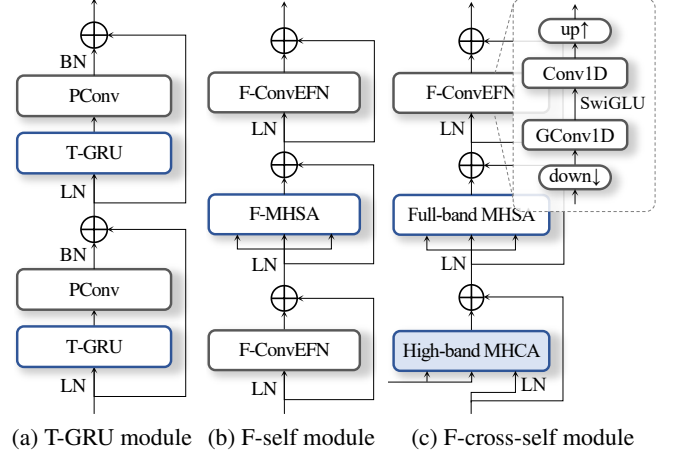


Fig. 4. Unit modules of TF-Refiner with the ConvEFN structure.

2.2. Asymmetric encoder-decoder

The TF-Encoder applies B_e blocks [16, 18], each an F-self module based on multi-head self-attention (MHSA) with rotary position encoding [27] and A_e heads followed by a T-GRU module (Figure 4), to $\mathbf{Z}_l$ and yields $\mathbf{Z}_e$. Band Merge prepares the complementary band as input-dependent queries through a residual convolution and normalization (Figure 3(b)), concatenates them with $\mathbf{Z}_e$ along frequency, and projects the channels from C_e to C_d to form $\mathbf{Z}_d$. The TF-Decoder applies B_d blocks, each an F-cross-self module followed by a T-GRU module. In the cross-self module only the F_h complementary positions query $\mathbf{Z}_e$ through multi-head cross-attention (high-band MHCA) with A_d heads; the low-band positions bypass it. A full-band MHSA with A_d heads over all F positions then relates the two bands. Both modules follow TF-Restormer [21] and use convolutional efficient feed-forward networks (ConvEFN) on a length-reduced sequence [28]: MHSA between two ConvEFNs in the F-self module, and high-band MHCA, full-band MHSA, and one ConvEFN in sequence in the F-cross-self module, with hidden widths H_e and H_d and two-group convolutions. The T-GRU module stacks two unidirectional GRU residual units.

2.3. Deep filtering

Let $X_{tf} \in \mathbb{C}$ denote the noisy STFT coefficient at frame t and bin f , and let $\mathbf{x}_{tf} \in \mathbb{C}^N$ collect its N neighboring coefficients in time and frequency. The Filter Head (Figure 3(c)) mixes the decoder channels with a pointwise convolution, restores the STFT frequency resolution with a transposed convolution along frequency, and estimates the complex filter tensor $\mathbf{W}$ with a convolutional layer. Following the three-component estimation structure of SR-CorrNet [29], three real vectors per bin $\mathbf{m}_{tf}, \mathbf{a}_{tf}, \mathbf{b}_{tf} \in \mathbb{R}^N$ give $\mathbf{w}_{tf} = \text{softplus}(\mathbf{m}_{tf}) \odot [\tanh(\mathbf{a}_{tf}) + j \tanh(\mathbf{b}_{tf})]$. As in deep filtering [25, 26], the enhanced STFT is $\hat{S}_{tf} = \mathbf{w}_{tf}^T \mathbf{x}_{tf}$, and the SFI-iSTFT reconstructs the waveform. The compressed features determine the filter, but the samples being combined are the uncompressed observations.

3. EXPERIMENTAL SETUP

3.1. Datasets and evaluation metrics

We evaluate on the 824-utterance VoiceBank+DEMAND (VBD) test set at 16 and 48 kHz [30]; its fixed mixtures use SNRs of

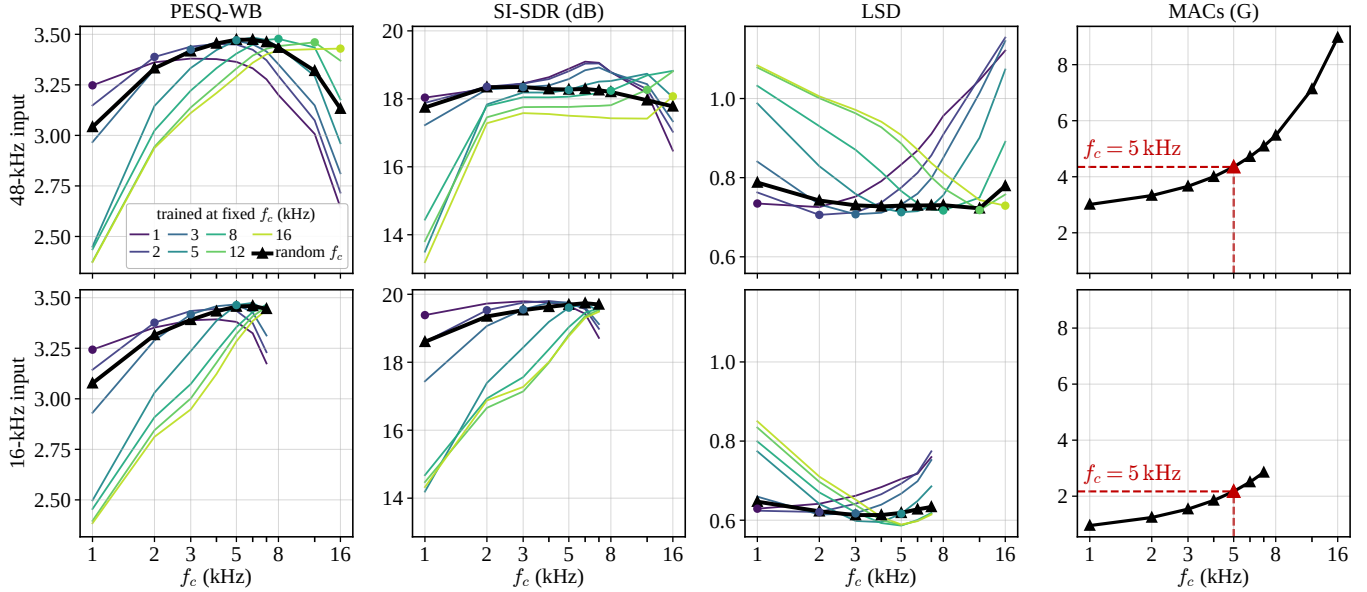


Fig. 5. Changing the analysis cutoff f_c at inference, for 48-kHz (top) and 16-kHz input (bottom). Colored curves: models trained with f_c fixed, one color per training cutoff, the circle marking that cutoff (capped at 7 kHz for 16-kHz input, hence no circle for the 8-, 12-, and 16-kHz models there). Black: the universal model trained with a randomly drawn cutoff. All curves come from one checkpoint per model evaluated at each f_c ; the cost depends only on the input rate and f_c , and red marks the default $f_c = 5$ kHz of Table 1.

Table 1. The universal model and the two rate-specific models on VBD at the default cutoff $f_c = 5$ kHz (*: rates absent from training).

Rate (kHz)	MACs	Model	PESQ [↑]	STOI [↑]	SI-SDR [↑]	LSD [↓]	LSD [↓] _{>8k}
16	2.17G	Noisy	1.97	0.921	8.47	1.193	—
		Universal	3.46	0.952	19.69	0.619	—
		16k only	3.35	0.950	19.20	0.624	—
		48k only*	3.03	0.928	18.08	0.804	—
24*	2.63G	Noisy	1.97	0.921	8.44	1.297	1.349
		Universal	3.44	0.952	19.62	0.651	0.560
		16k only	3.34	0.949	19.13	0.660	0.549
		48k only	3.18	0.939	18.58	0.758	0.702
32*	3.14G	Noisy	1.97	0.921	8.43	1.288	1.269
		Universal	3.47	0.953	19.47	0.660	0.595
		16k only	3.31	0.949	19.07	0.691	0.628
		48k only	3.29	0.946	18.60	0.725	0.691
44.1*	4.03G	Noisy	1.97	0.921	8.42	1.314	1.290
		Universal	3.47	0.952	18.27	0.721	0.705
		16k only	3.23	0.949	18.96	0.748	0.722
		48k only	3.34	0.949	17.61	0.751	0.752
48	4.35G	Noisy	1.97	0.921	8.41	1.314	1.285
		Universal	3.47	0.951	18.27	0.729	0.720
		16k only*	3.22	0.949	18.99	0.785	0.777
		48k only	3.35	0.949	17.74	0.735	0.730

{2.5, 7.5, 12.5, 17.5} dB. The 16-kHz data are the official 28-speaker VBD training pairs at {0, 5, 10, 15} dB, divided into 10,413 training and 1,159 validation utterances. The 48-kHz training data mix clean VCTK speech [31] from 98 training and seven validation speakers, excluding both test speakers, with DNS full-band noise [1] at an integer SNR drawn uniformly from {0, ..., 19} dB. VCTK utterances overlapping the VBD validation set are removed from the 48-kHz pool.

We report wideband PESQ [32], STOI [33], SI-SDR [34], and log-spectral distance (LSD) over the full input band and above 8 kHz (LSD_{>8k}, up to the input Nyquist frequency and hence undefined at

16 kHz). Inference runs at the input rate; SI-SDR and LSD are computed at that rate against the original clean signal, whereas PESQ and STOI are computed after resampling to 16 kHz. For the multi-rate evaluation the same 48-kHz test pairs are resampled to 16, 24, 32, and 44.1 kHz. We measure multiply-accumulate operations (MACs) for one second of input using ptflops.

3.2. Training and model configuration

We train the universal model by alternating 16-kHz VBD and 48-kHz VCTK+DNS batches. We use 20 epochs of 20,000 minibatches each (400,000 in total), batch size 2, 2-s crops, and AdamP [35] (weight decay 1e-2, $\beta = (0.95, 0.999)$). The learning rate is warmed up for 1,000 steps to 2e-3 and then cosine-decayed to 1e-6. Training draws cutoffs uniformly from {2, ..., 8, 12, 16} kHz for 48-kHz batches and {2, ..., 7} kHz for 16-kHz batches. One checkpoint per model is used for all reported input rates and inference cutoffs. We use the objective of FastEnhancer [7], $\mathcal{L} = 0.3\mathcal{L}_{\text{mag}} + 0.2\mathcal{L}_{\text{cplx}} + 0.3\mathcal{L}_{\text{cons}} + 0.2\mathcal{L}_{\text{wav}} + 0.001\mathcal{L}_{\text{pesq}}$, with power-compressed spectral terms. We use the TF-Refiner-B configuration in Table 3. The model applies deep filtering over a 3×3 TF neighborhood ($N = 9$: one past and one future frame, ± 1 bin), with 3×3 convolutions in Band Merge and filter prediction. Each convolution adds one future frame, giving a 40-ms look-ahead and, with the 40-ms window, an 80-ms algorithmic latency. The default cutoff is $f_c = 5$ kHz.

4. RESULTS

4.1. Universal enhancement across sampling rates

Table 1 evaluates the universal model trained at 16 and 48 kHz. The two rate-specific counterparts are trained only on data at their respective rate. The universal model maintains comparable perceptual quality at the unseen rates and improves every metric over the noisy

Table 2. Results on VBD at 16 and 48 kHz. Each TF-Refiner variant uses the same parameter set for both rates. GTCRN and UL-UNAS report SI-SNR. DeepFilterNet1/2 [10, 11] use DNS training data. *: zero-shot results for unseen rates. †: no-look-ahead version.

Method	Param	16 kHz				48 kHz			
		MACs	PESQ	SI-SDR	MACs	PESQ	SI-SDR	MACs	SI-SDR
Noisy	–	–	1.97	8.47	–	1.97	8.41	–	–
GTCRN [5]	0.02M	0.04G	2.87	18.83	–	–	–	–	–
UL-UNAS [6]	0.17M	0.03G	3.09	18.48	–	–	–	–	–
FastEnhancer-B [7]	0.09M	0.26G	3.13	19.0	–	–	–	–	–
FastEnhancer-S [7]	0.20M	0.66G	3.19	19.2	–	–	–	–	–
FastEnhancer-M [7]	0.49M	2.90G	3.24	19.4	–	–	–	–	–
TF-Refiner-B (16k)	0.47M	2.17G	3.35	19.20	4.35G	3.22*	18.99*	–	–
PercepNet [9]	8.00M	–	–	–	0.80G	2.73	–	–	–
DeepFilterNet [10]	1.78M	–	–	–	0.35G	2.81	16.63	–	–
DMF-Net [12]	7.84M	–	–	–	–	2.97	–	–	–
DeepFilterNet2 [11]	2.31M	–	–	–	0.36G	3.08	–	–	–
TF-Refiner-B (48k)	0.47M	2.17G	3.03*	18.08*	4.35G	3.35	17.74	–	–
TF-Refiner-B	0.47M	2.17G	3.46	19.69	4.35G	3.47	18.27	–	–
TF-Refiner-T†	0.09M	0.54G	3.20	19.07	1.27G	3.10	17.07	–	–
TF-Refiner-S†	0.19M	0.97G	3.37	19.42	2.03G	3.36	18.36	–	–
TF-Refiner-B†	0.45M	2.05G	3.30	19.42	3.85G	3.36	18.41	–	–

input. It obtains the best PESQ, STOI, and LSD at every rate and the best SI-SDR up to 32 kHz, whereas the 16-kHz model keeps the highest SI-SDR at 44.1 and 48 kHz, presumably because its training is confined to the power-dominant low band. The rate-specific models also transfer to unseen input rates without adaptation, although their PESQ decreases gradually. With the default cutoff of $f_c = 5$ kHz, the computational cost grows by about 2.0 times from 16 to 48 kHz, whereas the number of bins triples, because the encoder analysis band remains fixed while only the shallow decoder processes the wider observed band.

Because the VBD speech is drawn from VCTK, the two training streams do not provide disjoint clean-speech corpora; they differ mainly in sampling bandwidth and corruption generation. Combining VBD pairs with VCTK+DNS mixtures may therefore contribute to the universal model’s advantage, but the 48-kHz-only model already uses the larger VCTK+DNS pool yet remains below the universal model at its native rate. We note that Table 1 does not isolate the contributions of data diversity and joint sampling-rate exposure.

4.2. Configurable computation

Figure 5 compares the universal model with fixed-cutoff controls while changing f_c at inference for every model. The controls use the same 16/48-kHz training with f_c fixed at 1, 2, 3, 5, 8, 12, or 16 kHz. The total MACs increase with f_c because more positions enter the deep encoder. Fixed-cutoff models degrade as the inference cutoff moves far from their training cutoff. Random-cutoff training keeps the universal model close to the best fixed-cutoff model in PESQ and LSD over 2–8 kHz (2–7 kHz for 16-kHz input), while the fixed-cutoff models retain an advantage at the extremes.

Lowering f_c from the default 5 to 3 kHz reduces the cost by 29% at 16 kHz and 16% at 48 kHz with a small PESQ decrease. At lower cutoffs, the universal model also degrades more gradually than the model trained at a fixed 5-kHz cutoff. At 48 kHz, the model trained and evaluated at $f_c = 16$ kHz costs about twice as much as the universal model at the default cutoff without a higher PESQ.

Table 3. Model configurations of the TF-Refiner variants.

Model	Encoder				Decoder			
	B_e	C_e	H_e	A_e	B_d	C_d	H_d	A_d
TF-Refiner-T	4	24	42	4	2	12	24	2
TF-Refiner-S	4	32	88	4	2	16	48	4
TF-Refiner-B	4	48	144	4	2	24	72	4

Table 4. Ablation of the rate-specific 48-kHz model.

Model	PESQ†	STOI†	SI-SDR†	LSD↓	LSD↓ _{>8k}
TF-Refiner-B (48k)	3.35	0.949	17.74	0.735	0.730
w/o high-band MHCA	3.31	0.950	17.40	0.782	0.794
w/o Band Merge conv.	3.34	0.950	17.39	0.767	0.773
w/ direct mapping	3.11	0.944	15.95	0.801	0.812

4.3. Comparison with published systems

Table 2 compares published systems at their native rates with TF-Refiner variants, each using one parameter set at both rates. For single-rate comparisons, we include TF-Refiner-B (16k) and TF-Refiner-B (48k), the rate-specific models of Table 1. At the default $f_c = 5$ kHz, TF-Refiner-B (16k) achieves a PESQ of 3.35 versus 3.24 for FastEnhancer-M at 16 kHz, while its SI-SDR is 0.2 dB lower. TF-Refiner-B (48k) achieves a PESQ of 3.35 versus 3.08 for DeepFilterNet2 [11] at 48 kHz. The universal TF-Refiner-B further increases PESQ to 3.46 and 3.47 at 16 and 48 kHz, respectively.

The no-look-ahead variants (†) use frequency-only convolutions with kernel size 3 in Band Merge and filter prediction, with $N = 3$ same-frame frequency taps, giving a 40-ms algorithmic latency. TF-Refiner-B† also outperforms the published systems in PESQ at both rates. For scaling, we evaluate the tiny and small no-look-ahead variants using the configurations in Table 3. S† maintains comparable quality to B† at substantially lower cost, while T† trades some quality for further savings.

4.4. Ablation study

Table 4 ablates the modules that connect the analysis band to the full-band output. Removing high-band MHCA, through which the high-band queries attend to the encoder output, or the Band Merge convolution that forms those queries changes PESQ and STOI, which assess only the band below 8 kHz, by at most 0.04 but costs about 0.3 dB of SI-SDR and raises LSD_{>8k} from 0.730 to 0.794 and 0.773, respectively. Both modules thus mainly serve the observed high band. Replacing local filtering with direct complex mapping degrades every metric, lowering SI-SDR by 1.8 dB and raising LSD by 0.07, consistent with the filter preserving the fine structure of the observed STFT.

5. CONCLUSION

We presented TF-Refiner, a sampling-frequency-independent full-band speech enhancement model whose deep analysis is confined to a configurable band while a shallow decoder refines the observed remainder and filters the original noisy STFT. One universal model enhances 16- to 48-kHz inputs, including unseen rates, and the analysis cutoff selects a cost–quality operating point at inference without changing the architecture or retraining. Future work includes evaluation on unseen corpora and streaming latency measurement.

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